ASSIGNMENT

CO1 Integrated Engineering Analysis and Planning Assignment

Name :- Renganathan A

Reg. No. :- 192412533

Course code :- ECA5604

Course Name :- 5G/6G Communication

Engineering Design Report: Next-Stage Mobile Network Planning for a Regional Operator

1. Requirements and Approach

The operator needs a coherent plan for its next network stage. This report follows the required order: requirements, analysis, calculations, comparison, recommendation. It explains the evolution from 1G to the 6G vision, then analyses cell geometry and reuse, traffic and blocking, mobility and handoff, interference, spectrum bands and international coordination. It concludes with a phased deployment plan.

Planning data used. Hexagonal cells of radius r = 2 km; 19-cell planning cluster; reuse factor N = 7; 1,000 subscribers at 0.08 Erlang each; C = 100 channels with blocking target ≤ 2%; mobile speed 60 km/h; overlap dwell time 6 s; route cell length 1 km; handoff interruption 0.5 s; target-cell unavailability 1%; S = −70 dBm, I_1 = −85 dBm, I_2 = −90 dBm, SINR target 15 dB; c = 3×10^8 m/s.

2. Stage 1 – Evolution of Mobile Systems (1G to 6G)

Table 1. Generational comparison

Gen.	Main service capability	Access / radio technology	Key limitation	Motivated change
1G (AMPS)	Analog voice only	FDMA, 30 kHz channels, 800 MHz, 10 MHz allocation	Poor spectral efficiency, no security, no data, handset size	Digital transmission and better capacity
2G (GSM)	Digital voice, SMS, early data	TDMA, 200 kHz carrier, 8 slots, 900 MHz	Low data rate, circuit-switched voice focus	Wideband packet data
3G (UMTS)	Voice plus mobile broadband data, internet access	Wideband CDMA, 5 MHz carrier	Interference-limited capacity; modest rates for rich video	Higher rate, flexible spectrum use
4G (LTE)	All-IP mobile broadband, HD video	OFDMA with MIMO	Rate and latency not enough for URLLC and massive IoT; sub-6 GHz crowding	Wider bandwidth, diverse services
5G (NR)	eMBB, URLLC, massive IoT	Mid-band and mmWave, beamforming, flexible numerology	mmWave coverage is short and blockage-prone; dense deployment cost	Sensing, AI, THz and integrated services
6G (vision)	Holographic communication, tactile internet, ISAC, AI-native networking	THz bands, AI-native architecture, joint sensing and communication	Not standardised; THz range and hardware limits (vision only)	–

Calculations

Wavelength at 800 MHz: λ = c/f = $(3 \times 10^8)/(800 \times 10^6)$ = 0.375 m (37.5 cm).

Idealised number of 1G channels: N_ch = 10 MHz / 30 kHz = 10,000 kHz / 30 kHz ≈ 333 channels.

Practical qualification. This figure ignores guard bands, the duplex split and control channels. A real FDD analogue system needs a paired uplink and downlink for each call, so the usable number of simultaneous voice circuits is lower (about half if the 10 MHz is split between directions), and some channels carry signalling. Under reuse N = 7, only about 333/7 ≈ 47 channels would be available per cell.

2G improvement over 1G. Digital transmission allows voice coding, error control, encryption and more efficient modulation. GSM's TDMA divides each 200 kHz carrier into 8 slots, giving 8 users per carrier (200 kHz / 8 = 25 kHz per user, comparable to or better than 30 kHz per analogue user) with better quality, tolerance to interference (so tighter reuse is possible) and security. It also enabled SMS and low-rate data.

Synthesis

Across generations the role of spectrum shifted from simply dividing frequency among users (1G FDMA), to sharing by time (2G TDMA), by code (3G), and by frequency-time-space resources (4G OFDMA/MIMO and 5G beamforming). Capacity gains increasingly came from wider carriers, higher-order modulation, spatial reuse and densification, not only from more spectrum. Interference management moved from static frequency planning (reuse N = 7 or 12) toward dynamic coordination, power control and beam-level separation. By 5G and the 6G vision, spectrum spans low, mid and mmWave/THz bands, so propagation and cell density become design variables together with bandwidth.

3. Stage 2 – Cell Coverage and Frequency Reuse

Calculations

$$A_cell = (3\sqrt{3}/2)\, r^2 = 2.598 \times (2\ km)^2 = 10.39\ km^2$$

$$A_cluster = 19 \times 10.39\ km^2 = 197.45\ km^2$$

$$Reuse\ fraction = 1/N = 1/7 \approx 0.143\ (14.3\%\ of\ the\ channels\ per\ cell)$$

$$D = r\sqrt{(3N)} = 2\ km \times \sqrt{21} = 9.17\ km \quad (D/r = 4.58)$$

Table 2. Geometry and reuse results

Quantity	Result	Unit
Area of one hexagonal cell	10.39	km²
Area of 19-cell cluster	197.45	km²
Reuse fraction (N = 7)	0.143 (1/7)	–
Reuse distance D	9.17	km

Assumptions of the hexagonal model. It assumes flat terrain, omnidirectional antennas at cell centres, uniform propagation and identical power and cell size. It tiles the plane without gaps and gives a clean geometry for the reuse distance. Real cells are irregular because of terrain, buildings, shadowing and antenna patterns.

Geometric versus guaranteed radio coverage. A cell hexagon shows the nominal geometric area a site is intended to serve. Guaranteed radio coverage depends on received signal level, fading margin, indoor penetration, interference and noise at each location. A point inside the hexagon may still have poor service, and coverage can extend outside it.

Capacity-expansion techniques

- **Cell splitting:** reduces cell radius; more cells per area, so more channel reuse and capacity, but more sites, more handoffs and reuse geometry must be re-planned.
- **Sectoring:** directional antennas (e.g. 120°) reduce the number of co-channel interferers and improve C/I, which permits tighter reuse, but each sector has fewer channels (trunking loss) and inter-sector handoffs increase.
- **Microcells:** low-power street-level cells for hotspots and urban corridors; large capacity gain, lower coverage per site.
- **Picocells:** small indoor or hotspot cells (offices, malls, stations) that add capacity where demand is concentrated.
- **Femtocells:** very small user-installed indoor cells that offload macro traffic and improve indoor coverage, but need interference control and backhaul.

Recommendation. For the regional operator, start with 120° sectoring on existing sites, because the interference case in Stage 5 shows the 15 dB target is narrowly missed and sectoring directly reduces co-channel interference. Then use selective cell splitting and microcell/picocell overlays in the highest-demand areas. Effects: interference falls with sectoring but can rise with careless splitting; handoff frequency increases because users cross sector and smaller-cell boundaries more often.

4. Stage 3 – Traffic Capacity (Erlang B)

Model and inputs

Offered traffic is A = (number of subscribers) × (traffic per subscriber). Blocked calls are assumed cleared (Erlang B): Poisson arrivals, exponentially distributed holding times, C identical channels and no queueing. The recursion used is $B(A,0) = 1$; $B(A,k) = A \cdot B(A,k-1) / [k + A \cdot B(A,k-1)]$, $k = 1,\ldots,C$.

$$A = 1{,}000 \times 0.08\ Erlang = 80\ Erlang$$

Results

Table 3. Erlang B blocking for A = 80 Erlang

Channels C	Blocking probability	Meets 2% target?
80	8.41%	No
90	2.62%	No
91	2.25%	No
92	1.92%	Yes (smallest C)
100	0.40%	Yes (over-provided)

Interpretation. With 100 channels the blocking probability is about 0.40%, well below the 2% target. The smallest channel count that meets the target is C = 92 (B = 1.92%); with 91 channels the blocking is 2.25%. Therefore one 100-channel base-station group serves the area, with about 8 spare channels (and 12.5% of the channel group unused at the target grade of service). ⌈92/100⌉ = 1 channel group.

Why this is only first-order. The calculation treats the area as a single pooled traffic source, ignores the spatial distribution of users, reuse constraints (channels cannot all be used in every cell), control channels, sectoring, handoff traffic, peak-to-average variations and growth. A site plan needs coverage, per-cell traffic maps, backhaul and interference analysis.

Trade-offs. Adding channels lowers blocking sharply at first, then with diminishing returns, since each extra channel costs spectrum and equipment. Running at 92 channels is efficient (about 87% utilisation of carried load: 80 × (1−0.0192)/92) but leaves little margin for traffic growth; the 100-channel design gives headroom at modest extra cost.

Assumption. Subscribers' busy-hour activity is independent and uniformly 0.08 Erlang (about 4.8 minutes per busy hour), and 100% of the demand is served by this one base station.

5. Stage 4 – Mobility and Handoff

Calculations

$$v = 60\ km/h = 16.67\ m/s$$

$$Distance\ in\ overlap\ interval = v \times t = 16.67\ m/s \times 6\ s = 100\ m$$

$$Time\ to\ cross\ a\ 1\ km\ cell = 1000\ m\ /\ 16.67\ m/s = 60\ s$$

$$Hard\ handoffs\ per\ minute \approx 60\ s\ /\ 60\ s = 1\ handoff\ per\ minute$$

Interpretation of the 1 km length. The 1 km is taken as the nominal route distance between successive handoff points (one cell crossing per 1 km), so each cell is crossed in 60 s. The true rate would differ for the 2 km radius cells in Stage 2 (a path through a cell centre is about 4 km, giving about one handoff per 4 minutes); this shows how sensitive the result is to route geometry.

Table 4. Handoff types

Type	Operation	Continuity	Resource use
Hard	Break-before-make; old link released before new link set up	Brief interruption (0.5 s here)	Lowest; one link at a time
Soft	Make-before-break; simultaneous links to two cells (CDMA-type systems)	Seamless	Higher; two links and network resources
Softer	Soft handoff between sectors of the same base station	Seamless	Moderate; combining done at the base station

Call-drop risk estimate

The handoff interruption of 0.5 s fits within the 6 s overlap dwell time, so the system has time to attempt access. If the target cell is unavailable (no free channel) with probability 1%, a hard handoff fails with probability about 0.01, so the simple drop risk is about 1% per handoff. At 1 handoff per

minute this is roughly 1% per minute of call time; for a 5-minute call the probability of at least one drop is $1 - 0.99^5 \approx 4.9\%$. The 0.5 s interruption itself is about 0.83% of each 60 s cell crossing (0.5/60), which is a quality hit rather than a drop.

Limits. The estimate assumes independent handoffs, a constant 1% unavailability, constant speed on a straight route, correct signal-level triggers and no ping-pong, no channel reservation (guard channels), and does not model radio failures such as fading, late triggers or a poor target-cell signal.

6. Stage 5 – Interference and Spectrum-Band Comparison

Interference analysis

$$C/I \text{ (co-channel only)} = S - I_1 = -70 - (-85) = 15 \text{ dB}$$

Linear powers (P = 10^(dBm/10) mW):

$$I_1 = 10\text{^}(-8.5) = 3.162 \times 10^{-9} \text{ mW};\ I_2 = 10\text{^}(-9.0) = 1.000 \times 10^{-9} \text{ mW}$$

$$I_total = 4.162 \times 10^{-9} \text{ mW} = -83.81 \text{ dBm}$$

$$SINR = S - I_total = -70 - (-83.81) = 13.81 \text{ dB}$$

Table 5. Interference results

Quantity	Value	Target / outcome
C/I with co-channel only	15.00 dB	Just equals 15 dB
Total interference ($I_1 + I_2$)	−83.81 dBm	–
SINR with both interferers	13.81 dB	Fails by 1.19 dB

Decision. The 15 dB target is not met once adjacent-channel interference is included; the link goes from exactly meeting the target to 1.19 dB short. Thermal noise is not given, so SINR here is effectively the signal-to-interference ratio; including noise would reduce it slightly.

Recommended mitigation. Use 120° sectoring (with antenna down-tilt), which cuts the number of co-channel interferers seen by each sector, and tighten adjacent-channel filtering/guard spacing. Reducing I_1 is the most effective step because it is the dominant term (about 76% of total interference). A reduction of the total interference by more than 1.2 dB is needed. Trade-off: more antennas and equipment, smaller sector channel pools (trunking efficiency loss) and more sector handoffs. Increasing the reuse cluster size (for example N = 7 to 12) is an alternative, but it reduces the channels per cell by about 42%.

Spectrum-band comparison at d = 100 m

$$FSPL(dB) = 20 \log_{10}(4\pi d/\lambda),\ \lambda = c/f$$

Table 6. Wavelength and free-space path loss at 100 m

Band	Wavelength	FSPL at 100 m	Relative to 900 MHz	Typical use
900 MHz	0.333 m (33.3 cm)	71.53 dB	0 dB (reference)	Wide-area coverage
3.5 GHz	0.0857 m (8.57 cm)	83.32 dB	+11.80 dB	5G mid-band

Band	Wavelength	FSPL at 100 m	Relative to 900 MHz	Typical use
28 GHz	0.0107 m (1.07 cm)	101.38 dB	+29.86 dB	5G mmWave hotspots

Table 7. Deployment implications

Band	Penetration	Antenna size	Coverage / capacity
900 MHz	Best; good building penetration	Largest elements	Largest cells, best coverage, limited bandwidth so lower capacity
3.5 GHz	Moderate	Medium; practical massive MIMO arrays	Balanced coverage and capacity; ideal for 5G eMBB
28 GHz	Poor; blocked by walls, foliage and bodies	Very small elements; large arrays in a small area	Short range, very high capacity from wide bandwidth, needs dense small cells and beamforming

Why path loss alone does not decide performance. The free-space formula assumes isotropic antennas. Higher frequencies allow many more antenna elements in the same aperture, so beamforming gain can offset a large part of the extra loss. Achievable throughput also depends on available bandwidth (far larger at 28 GHz), SINR, interference, line-of-sight conditions, transmit power, noise bandwidth, network density and traffic demand. A high-loss band with wide bandwidth and dense small cells can give higher capacity than a low-loss narrowband carrier.

7. Stage 6 – Spectrum Coordination and Deployment Recommendation

International spectrum coordination

- **ITU-R:** the ITU Radiocommunication Sector, which manages the global radio-frequency spectrum and satellite orbits, develops technical recommendations and the international Radio Regulations.
- **WRC (World Radiocommunication Conference):** held roughly every 3–4 years; revises the Radio Regulations and decides frequency allocations (for example identifying bands for IMT/5G).
- **Radiocommunication Bureau (BR):** the ITU-R's executive arm; records frequency assignments in the Master International Frequency Register, processes notifications, supports coordination and prepares conference material.

Effect on national regulators and operators. WRC allocations and ITU rules guide national band plans and licensing. Harmonised bands (for example 3.5 GHz) lead to economies of scale in devices and equipment and to lower cross-border interference, whereas the operator must follow the national regulator's licence conditions, power limits and border coordination agreements.

Recommended phased strategy

Table 8. Phased deployment plan

Phase	Action	Supporting evidence from analysis
1	Re-use 900 MHz macro sites; deploy 120° sectoring and tune power and tilt; provision about 92–100 channels per site	Erlang B: 92 channels meet 2% blocking (100 gives 0.40%). SINR of 13.81 dB misses 15 dB, so interference reduction is needed first.
2	Add 3.5 GHz 5G NR with massive MIMO on macro sites for capacity (eMBB)	Path loss is +11.8 dB versus 900 MHz, but wider bandwidth and beamforming compensate; moderate penetration.
3	Selective cell splitting; micro/picocells and 28 GHz hotspots in dense, high-demand areas	28 GHz has +29.9 dB path loss, so it suits only short-range hotspots; smaller cells raise capacity but increase handoffs.
4	Prepare for 6G features (ISAC, AI-based optimisation) as standards mature	Vision-stage technologies; keep the architecture upgradable.

Integration. The evolution from analogue to OFDMA/MIMO shows that capacity is gained from reuse, bandwidth and spatial processing. The cell plan (A_{cell} = 10.39 km², D = 9.17 km) sets the reuse geometry; the Erlang B result sets the channel dimensioning; handoffs at about 1 per minute with about 1% failure risk require reserved handoff capacity and well-set margins; the interference result requires sectoring or equivalent mitigation; and the band comparison puts coverage on 900 MHz, capacity on 3.5 GHz and hotspots on 28 GHz.

Assumptions and uncertainties to verify before deployment

- Actual terrain, clutter and measured propagation (versus free-space and hexagonal assumptions); shadowing and fading margins.
- Real traffic distribution, growth, and busy-hour profile versus the uniform 0.08 Erlang assumption; data traffic is not modelled by Erlang B.
- Noise power, antenna patterns, true interferer count and power control; SINR margin is only 1.19 dB away from the target.
- Route geometry and actual handoff rates and failure probability; handoff triggers and ping-pong effects.
- Spectrum availability, licence terms, national band plans and cross-border coordination with neighbouring countries.
- Cost, backhaul, site availability and equipment/device ecosystem for each band.

References

- T. S. Rappaport, Wireless Communications: Principles and Practice, 2nd ed., Prentice Hall, 2002.
- A. Goldsmith, Wireless Communications, Cambridge University Press, 2005.
- ITU-R, Radio Regulations and IMT recommendations (International Telecommunication Union).
- Course assignment planning data supplied for ECA5604 (CO1).